Phase 2 Report

By James Hoang, Soumil Makhija, Sunny Machhiana and Noah Cooper

For implementing all the features, we came up with in the design phase we decided to split the work into 4 parts.

Person 1 (Soumil) was responsible for the games entities such as the Player and the enemy Drones, the items (med kits and keys) as well as some of the utility objects like the resource loader.

Person 2 (Noah) was responsible for the map and locations such as Walls, Lockers, the Alarm System and the traps as well.

Person 3 (James) was responsible for the UI and the game rendering working on parts like the display for how many key fragments were collected and displaying how much health the player has.

Person 4 (Sunny) was responsible for a lot of the controller and builder objects such as the Game Factory object and the Input handling as well as the Enums used throughout the game's code.

During the course of programming the game we did make some changes due to difficulties we encountered. We realised the scope of the ideas we had come up with was a bit too much to do before the deadline, so we cut some of our extra features like the save/load system and procedural generation. We also changed how some of the traps work in order to avoid making the objects all too dependent on each other resulting in "spaghetti code".

We used JavaFX to do the rendering and UI because after some research James determined that it would work well for the type of game we are making, and it would make it easier to implement the graphics.

Some of the biggest challenges we faced were getting all parts of the game (engine, UI, and map) to work together smoothly and update in sync without causing bugs or timing issues and getting all the map objects to interact with the player and drones without needing to couple the player and map objects together too much.